
POMODORO TIMER WEB APPLICATION

A FLEXISAF T2 2025 FINAL
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Introduction

What is a Pomodoro Timer?

A time management technique (25 mins work + 5 mins break).

Boosts productivity by dividing work into intervals.

Project Goal:

Build a customizable, user-friendly timer with React.js.



Key Features

Customizable Timer:

Adjustable work/break durations (sliders in settings).

Session Tracking:

Tracks completed work sessions and suggests long breaks.

Visual Feedback:

Color-coded sessions (work/break) and progress bar.

Responsive UI: Works on mobile and desktop.

Technologies Used



Frontend: React.js(for logic), Tailwind CSS(Styling and responsive design).



Icons: React Icons.



Audio: HTML5 Audio AP for session end alerts.

Timer Architecture

Timer Component (Parent): Manages state (time, sessions).

Settings Component (Child): Receives props for durations and callbacks.

Data Flow: Parent-to-child via props (e.g. workDuration).

Child-to-parent via callbacks (e.g., onWorkDurationChange).

Code Snippet (Timer.jsx)

State Management

```
const [time, setTime] = useState(1500); // 25 mins in seconds
const [isRunning, setIsRunning] = useState(false);
```

Countdown Logic

```
useEffect(() => {
  if (isRunning && time > 0) {
    const timer = setInterval(() => setTime(t => t - 1), 1000);
    return () => clearInterval(timer);
  }
}, [isRunning, time]);
```

Code Snippet (Settings.jsx)

Slider Input

```
<input
  type="range"
  id="work-duration"
  min="1"
  max="60"
  step="1"
  value={workDuration / 60}
  onChange={onWorkDurationChange}
  className="w-full h-2 bg-gray-200 rounded-lg appearance-none cursor-pointer"
/>
```

Props Flow

```
<Settings
  workDuration={workSession}
  onWorkDurationChange={handleWorkDurationChange}
  isOpen={settingsOpen}
  onClose={closeSettings}
/>
```

Challenges & Solution

Challenge: Syncing timer with dynamic durations.



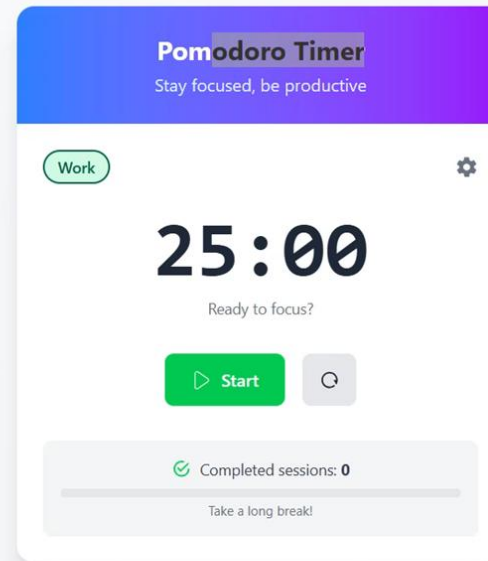
Solution: useEffect dependency array.



Challenge: Prop drilling.



Solution: Lifted state to parent (Timer.jsx).



Check [Github Repo](#)

Demo

This is the timer in action showing starting, pausing, and settings adjustment. Also includes work, short break and long break sessions.

Conclusion



Built a customizable Pomodoro Timer with React + Tailwind CSS.



Learnings: State management, component composition, UI/UX principles.

THANK YOU

